Updated Rules — Original vs. LLM-Generated (Scenario 2, LLM-consensus only)

Signal	Direction	Behavior	Manual	GPT-o4-mini	Sonnet-4	GPT-5.1
M0 Macrophage						
Pro-Inflammatory Factor	Increases	Transform To M1 Macrophage	○	○	●	●
Anti-Inflammatory Factor	Increases	Transform To M2 Macrophage	○	○	●	●
Substrate Gradient: Apoptotic Debris	Increases	Chemotactic Response To Substrate	○	●	○	○
M1 Macrophage						
Pro-Inflammatory Factor	Increases	Chemotactic Response To Pro-Inflammatory...	○	○	●	●
Substrate Gradient: Necrotic Debris	Increases	Phagocytose Necrotic Cell	○	●	○	○
Contact With Malignant Epithelial Cell	Increases	Attack Malignant Epithelial Cell	○	○	●	○
Apoptotic Debris	Increases	Apoptotic Debris Uptake	○	○	●	○
Pro-Inflammatory Factor	Increases	Attack Malignant Epithelial Cell	○	○	○	●
M2 Macrophage						
Substrate: Anti-Inflammatory Factor	Increases	Migration Speed	○	●	○	○
Contact With Malignant Epithelial Cell	Decreases	Migration Speed	○	○	●	○
Apoptotic Debris	Increases	Apoptotic Debris Uptake	○	○	●	○
Anti-Inflammatory Factor	Increases	Chemotactic Response To Anti-Inflammatory...	○	○	○	●
Anti-Inflammatory Factor	Increases	Phagocytose Apoptotic Cell	○	○	○	●
Effector T Cell						
Anti-Inflammatory Factor	Increases	Transform To Exhausted T Cell	●	○	●	●
Pro-Inflammatory Factor	Increases	Attack Malignant Epithelial Cell	●	○	○	●
Pro-Inflammatory Factor	Increases	Migration Speed	●	○	●	○
Pro-Inflammatory Factor	Increases	Chemotactic Response To Pro-Inflammatory...	○	○	●	●
Substrate: Pro-Inflammatory Factor	Increases	Attack Cell Type Malignant Epithelial Cell	○	●	○	○
Contact With Malignant Epithelial Cell	Increases	Attack Malignant Epithelial Cell	○	○	●	○
Oxygen	Increases	Attack Malignant Epithelial Cell	○	○	○	●
Exhausted T Cell						
Anti-Inflammatory Factor	Decreases	Migration Speed	●	○	●	○
Anti-Inflammatory Factor	Decreases	Attack Malignant Epithelial Cell	○	○	○	●
Malignant Epithelial Cell						
Oxygen	Decreases	Necrosis	●	○	○	●
Oxygen	Increases	Cycle Entry	●	○	○	●
Oxygen	Decreases	Migration Speed	●	○	●	○
Substrate: Oxygen	Increases	Cycle Entry	○	●	○	○
Contact With Effector T Cell	Increases	Pro-Inflammatory Factor Secretion	○	○	●	○
Oxygen	Decreases	Cycle Entry	○	○	●	○